

JOSE LUIS LOPEZ VEGA

Mechatronic Engineer | Manufacturing & Maintenance Specialist

Tijuana, Baja California | 667-343-0490 | jolulove0@gmail.com

PROFESSIONAL SUMMARY

Result-oriented Mechatronic Engineer with over 5 years of experience in manufacturing and industrial maintenance. Proven track record in New Product Introduction (NPI), process validation (IQ/OQ/PQ), and operational efficiency improvements within the medical device industry. Expert in 3D modeling, Lean Six Sigma methodologies, and cross-functional team leadership. Bilingual (English/Spanish) and committed to continuous technical advancement.

TECHNICAL SKILLS

- **Design & Engineering:** SolidWorks (Advanced), AutoCAD, DFM & DFA, Layout Development, Floor Layout Optimization.
 - **Quality & Compliance:** IQ/OQ/PQ Protocols, PFMEA/SFMEA, PPAP, 8Ds, 5 Whys, Root Cause Analysis, EU MDR Regulation Compliance.
 - **Operational Control:** Lean Six Sigma, Process Control, Agile MAP, Minitab, OEE Analysis.
 - **Software & Systems:** SAP, Microsoft Office Suite, PLC Basics, Hydraulic & Electric Systems.
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PROFESSIONAL EXPERIENCE

INDUSTRIAL COM | Maintenance Engineer | 2025 – 2026

- Direct and coordinate the implementation of large-scale industrial installation projects and technical services.
- Lead incident response teams, optimizing task reassignment to improve operational efficiency by minimizing downtime.
- Supervise compliance with the Master Maintenance Plan (Preventive/Corrective), ensuring 100% equipment availability.
- Provide specialized technical advice to key clients regarding equipment performance and system upgrades.

MEDTRONIC | Manufacturing Engineer (Coronary Products) | 2022 – 2024

- Executed New Product Introduction (NPI) strategies, focusing on machinery implementation and cleanroom distribution / Floor Layout Optimization.
- Ensured strict compliance with international medical device standards, including EU MDR regulations.
- Monitored and enhanced production metrics through continuous Overall Equipment Effectiveness (OEE) analysis.
- Developed and executed IQ, OQ, and PQ qualification protocols for new production lines (Catheter Line, Amphirion).

- Applied Lean Six Sigma methodologies to optimize production processes and reduce scrap rates.
- Utilized SAP for production control, inventory management, and documentation workflow.
- Optimized manufacturing through the design of specialized fixtures and tooling modifications in Solidworks.
- Collaborated with global suppliers to ensure technical specifications met quality standards and project timelines.
- Managed training and certification for operational personnel on new instruction sheets and safety protocols.

MEDTRONIC | *Engineering Intern (Design Department)* | 2020 – 2022

- Developed 3D industrial models for machinery and virtual cleanroom environments to enhance facility planning.
- Programmed virtual training modules, reducing onboarding time for new operational staff.
- Analyzed and updated manufacturing instruction sheets (MI) to ensure alignment with current production standards.
- Implemented security measure improvements for cleanroom access control.

EDUCATION

INSTITUTO TECNOLÓGICO DE HERMOSILLO (ITH)

Bachelor of Engineering in Mechatronics | 2015 – 2020

Specialty: Automotive and Aerospace Design

LANGUAGES

Spanish: Native

English: Advanced (B2 Level - Cambridge English Test / ITH University)